Abd Elrahman Elgarhy

Senior Electrical Engineer

Riyadh City, Saudi Arabia I (+966)581380118 I Abdooelgarah@gmail.com

Linked in. [https://www.linkedin.com/in/eng-abdelrahman-elgarah-1343aab3/]

Google Scholar: A.Elgarhy - Google Scholar

PROFESSIONAL SUMMARY

Senior Electrical Engineer with nearly 10 years of experience in renewable energy, power systems, ETAP studies, testing & commissioning, smart grids, electrical design, and project management across industrial and utility-scale projects in Saudi Arabia and Egypt.

Experienced in leading HV/MV/LV power system studies including Load Flow, Short Circuit, Arc Flash, Protection Coordination, Harmonic Analysis, and Battery Sizing using ETAP software. Strong background in renewable energy systems, solar PV design, substations, protection relay testing, and O&M activities.

Worked on major projects with ARAMCO, SIPCHEM, KAUST, SEC, and petrochemical facilities. strong technical, analytical, and leadership capabilities.

CORE COMPETENCIES

- Renewable Energy Systems
- Solar PV Design & Simulation
- ETAP Power System Studies
- Testing & Commissioning
- Protection Relay Testing
- Arc Flash & Coordination Studies
- Harmonic Analysis (IEEE 519)
- Battery Sizing Calculations
- HV/MV/LV Systems
- Power Quality Analysis
- Smart Grid Systems
- Project Management
- O&M Activities
- Electrical Design & Shop Drawings
- MATLAB/Simulink
- AutoCAD & DIALux
- PLC & Automation

TECHNICAL SKILLS

Software

- ETAP
- MATLAB/Simulink
- PVsyst
- PSCAD
- AutoCAD 2D
- PV CAD
- BESS
- DIALux
- LabVIEW
- Arduino

Engineering Studies

- Load Flow Analysis
- Cable sizing
- Short Circuit Analysis
- Arc Flash Analysis
- Protection Coordination Studies
- Transient Stability
- Motor starting
- Harmonic Analysis
- Battery Sizing
- Power Quality Studies

Testing & Commissioning

- CT/VT Testing
- Relay Testing
- Scheme Check
- UPS & Battery Testing
- Switchgear Testing

PROFESSIONAL EXPERIENCE

❖ **Future Technologies Ltd – Saudi Arabia**

Senior Electrical / Project Engineer

2022 – Present

Key Projects & Responsibilities

ARAMCO – Riyadh Refinery

Project Manager / ETAP Modeling Engineer

- Led power system studies for 23 substations.
- Performed Arc Flash, Short Circuit, and Protection Coordination studies using ETAP.
- Updated and validated ETAP models for refinery substations.
- Prepared technical reports and engineering documentation.
- SIPCHEM – Petrochemical Complex

Project Manager / Group Leader

- Managed field engineering and data collection activities for 17 substations.
- Verified Single Line Diagrams and protection device data.
- Conducted power quality and harmonic measurements using FLUKE 435.
- Coordinated engineering teams during site activities.
- KAUST – King Abdullah University of Science and Technology

Project Engineer

- Performed complete electrical system design for new substations.
- Conducted Load Flow, Short Circuit, Arc Flash, and Coordination studies.
- Supported power system modeling and simulation activities.

SABTANK

Project Engineer

- Performed battery sizing calculations and simulations.
- Conducted testing & commissioning activities.
- Prepared final technical reports.

Saudi Electricity Company – Plant 10

- Testing & Commissioning Engineer
- Performed testing and commissioning for 13.8 kV switchgear.
- Conducted CT, VT, transformer, and protection relay testing.

ARAMCO – Jeddah Terminal

Project Engineer

- Performed power system modeling and simulation for substations.

TASNEE Petrochemical Factory

Project Engineer

- Managed installation of new Marshalling Panel Cabinet in PPX Substation.
- Led testing and commissioning activities.

❖ **Alhasm Contracting – Saudi Arabia**

MEP Project Engineer

2020 – 2022

- Managed MEP site activities and technical office operations.
- Planned, executed, and handed over multiple MEP projects.
- Coordinated multidisciplinary engineering teams.
- Supervised project schedules and technical documentation.

❖ **Egyptian Swedish Co. for Trading Engineering Contracting – Egypt**

Electrical Engineer

2018 – 2020

- Managed maintenance activities for UPS systems and electrical equipment.
- Improved preventive maintenance programs and inventory systems.
- Conducted troubleshooting and repair for electrical systems.
- Performed testing & commissioning for solar PV stations.
- Installed and maintained solar energy systems.

❖ **Kandil Steel – Egypt**

Maintenance Engineer

2016 – 2018

- Managed maintenance programs for industrial production lines.
- Reduced downtime through preventive maintenance strategies.
- Modified PLC and SCADA systems according to production requirements.
- Designed power management systems using LabVIEW and Schneider modules.
- Conducted machine upgrading and automation projects.

Academic experiences

❖ **Zagazig University – Egypt**

Instructor & Research Assistant

2016 – 2018

- Instructor for Electrical Circuits, Power Electronics, Protection Systems, Renewable Energy, and Solar Energy.
- Instructor in robotics, automation, classic control, and electronic board design.
- Conducted training in Arduino, LabVIEW, and PLC programming.
- Participated in robotics and smart home projects.

IEEE Activities

- IEEE Member
- Vice Chairman IEEE ZSB – 2014
- Chairman IEEE RAS ZSC – 2015 & 2016

Education

❖ **Master's Degree in Mechatronics**

Zagazig University & Bochum University (Germany) 2022

- GPA: 3.24 / 4

❖ **Bachelor's degree in electrical Power & Machines Engineering**

Zagazig University – Egypt 2016

- Final Grade: Good
- Graduation Project: Smart Home using LabVIEW & myRIO
- Graduation Project Grade: Excellent

Research area

❖ **RESEARCH INTERESTS**

- Renewable Energy Systems
- Solar & Wind Energy
- Smart Grid Systems
- Electric Vehicles
- Energy Management Systems
- MATLAB/Simulink Modeling
- Hybrid Energy Systems

Master research

- Integration of fuel cells into an off-grid hybrid system using wave and solar energy.
- Simulation model of hybrid renewable sources integration using MATLAB/Simulink.

under review paper

- Impact of Electric Vehicle Charging Load Topology and Storage-Assisted Operation on Transient Stability of Renewable-Integrated Distribution Grids: A Comparative Dynamic Study

publications

[1] M. Talaat, **A. Elgarhy**, M. H. Elkholy, and M. A. Farahat, "Integration of fuel cells into an off-grid hybrid system using wave and solar energy," International Journal of Electrical Power & Energy Systems, vol. 130, p. 106939, 2021/09/01/ 2021, https://doi.org/10.1016/j.ijepes.2021.106939, **impact factor : 3.588.**

[2] **A. Elgarhy**, M. Talaat, M. J. T. E. I. J. o. E. S. Farahat, and Technology, "Simulation Model of Hybrid Renewable Sources Integration Using MATLAB/Simulink," vol. 37, no. 2, pp. 39-48, 2022.
DOI: 10.21608/EIJEST.2021.86559.1081

[3] M. Talaat, A. Alblawi, M. Tayseer, and M. H. Elkholy, "FPGA control system technology for integrating the PV/wave/FC hybrid system using ANN optimized by MFO techniques," Sustainable Cities and Society, vol. 80, p. 103825, 2022/05/01/ 2022,
https://doi.org/10.1016/j.scs.2022.103825. **impact factor: 7.587 (Participated in this paper)**

[4] M.H. Elkholy, Tomonobu Senjyu, Mohammed Elsayed Lotfy, **A. Elgarhy**, Nehad S. Ali and Tamer S. Gaafar"Design and Implementation of a Real-Time Smart Home Management System Considering Energy Saving," Advanced Renewable Energy for Sustainability, 2022, 14, 13840. DOI: 10.3390/su142113840. **impact factor: 3.889**

[5] S. M. Abd Elazim, M. H. Elkholy, **A. Elgarhy**, T. Senjyu, M. M. Gamil, D. Song, E. S. Ali,G. A. Ludin,A. S. Irshad, and M. E. Lotfy, "Enhancing stability and power quality in electric vehicle charging stations powered by hybrid energy sources through harmonic mitigation and load management," Scientific Reports, vol. 15, no. 28077, 2025, DOI: 10.1038/s41598-025-75914-0.

Internet research

Google Scholar	https://scholar.google.com/citations?user=HhwCDMgAAAAJ&hl=ar
ORCID	https://orcid.org/my-orcid?orcid=0000-0002-1692-1942
Research gate	https://www.researchgate.net/profile/A-Elgarhy

Training/Courses/Certificates

Training

The Technology of Manufacturing Electric cables and quality control on all production steps in July 2012 at Elsweedy Company July 2012
Abu Zaabal Company for Engineering Industries (military Production 100) 1/7/2013 to 31/7/2013
Misr contraco Co Group in Distribution system 1/8/2013 to 1/9/2013
Misr petroleum in Alexandria 1/7/2014 to 1/9/2014
safety orientation training in Aramco (Al Riyadh Refinery)
safety orientation training in SIPCHEM

Courses

- Electrical power system distribution basic application.
- Electric installation design from Engineers Syndicate.
- Electric cables and all production steps at Energy power cable.
- Classic control from AC (Automatic control) system included temperature control.
- PLC (programmable logic controller), SIEMENS, s7_300
- Microcontroller programing includes (Proteus, LabVIEW, PIC)
- Arduino.

Other Particulars

- **Full Name:** Abdelrahman Mohamed Garah Garhy.
- **Date of Birth:** Aug.1st, 1991.
- **Nationality:** Egyptian.

Languages

- **Arabic: Native.**
- **English: Very Good.**